



BusRes

Sign in to your account

Select Role

☒ ADMIN ☐ MANAGER ☐ PASSENGER

Username

Maheen

Password

•••••



Sign In

New Passenger? Create an account below:

+ Create Account



Bus Seat Reservation System

Data Structures & Algorithms Lab

Group Members:

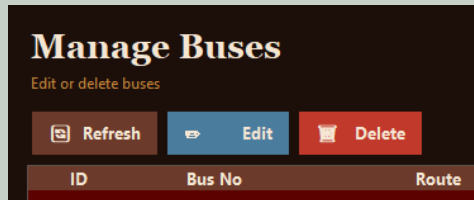
Maheen Akbar · Kinza batool

Course Details

Instructor: Engr. Faisal Zia · BSE-4A · Spring 2026

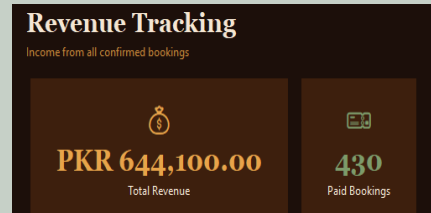
PROJECT OBJECTIVES

The Bus Seat Reservation System is a desktop database application built with a python tkinter frontend and Microsoft SQL Server Backend. It supports end-to-end booking operations: creation, booking_payments, promo_codes and waitlist managements



- **Manage Buses**

- Create, update and categorize buses with date/time, capacity, venue.



- **Handle Bookings**

Real Time Booking Flow with seat counts ,booking statuses and history tracking



- **Process Payments**

- Secure payment Integration , transaction logging, and PDF Ticket/ Invoice generation



User Roles & Capabilities

Role-based access ensures secure, tailored experiences

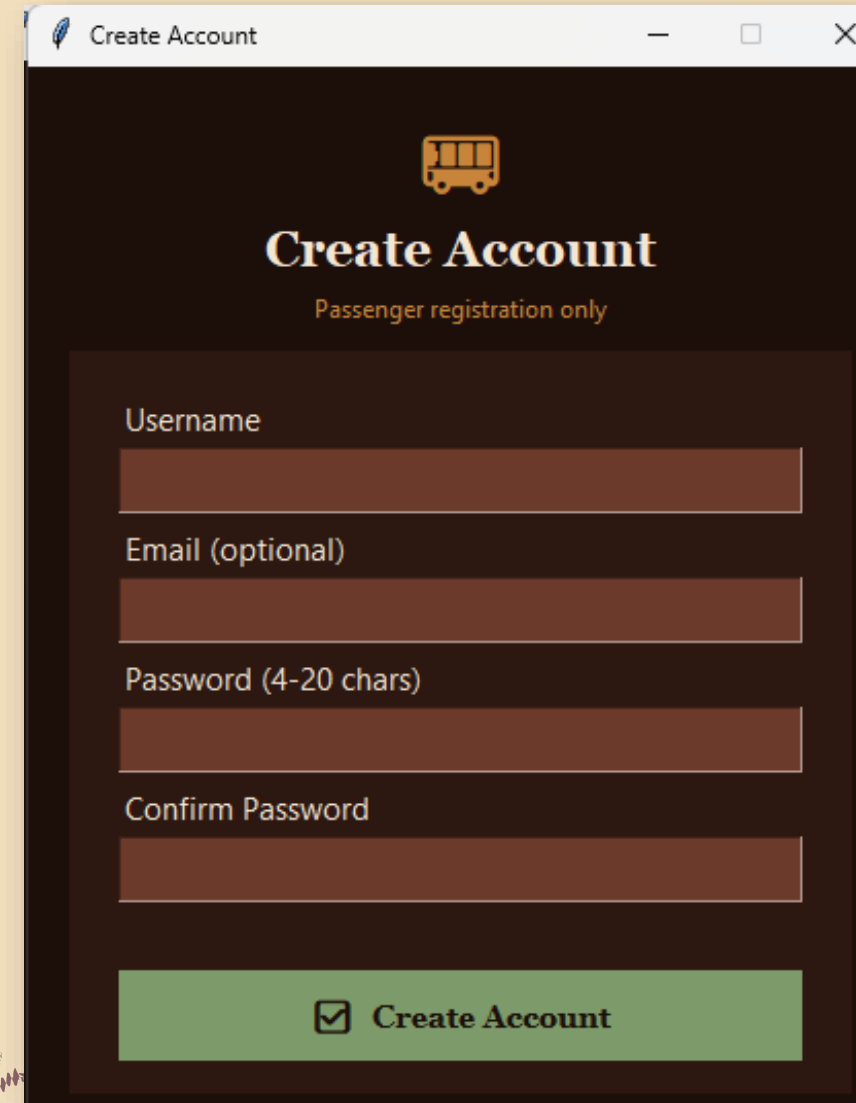
Admin:

Full systems control---user management, analytics access, global settings

Manager:

Bus Creation, schedule update, manage bookings and promo codes

Passenger: Browse Buses, make bookings, redeem promo_codes, view tickets.



Create Account



Create Account

Passenger registration only

Username

Email (optional)

Password (4-20 chars)

Confirm Password

☒ Create Account

Flow

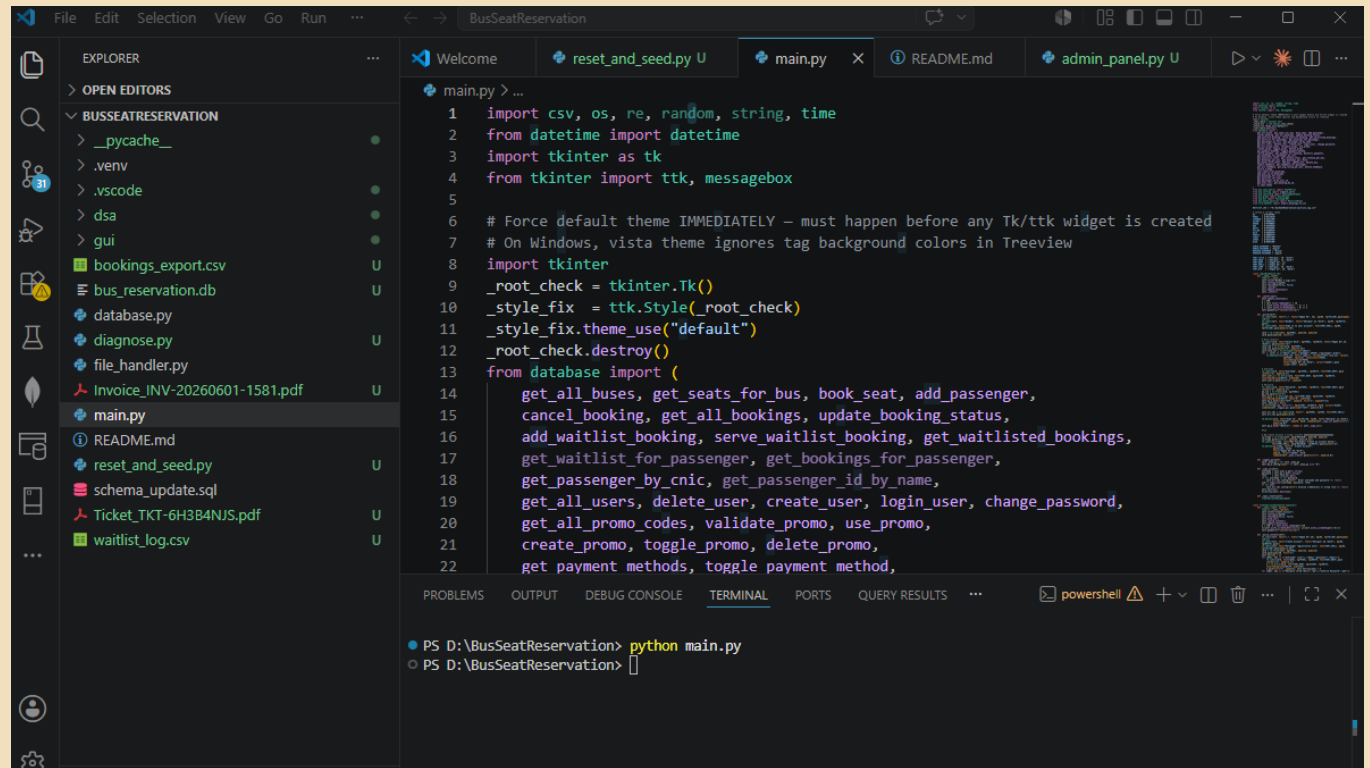
User → Tkinter GUI → pyodbc Connector → Microsoft SQL Server

Components:

Frontend: Python Tkinter (Desktop UI)

Backend: SQL Server with normalized relational schema

Connector: pyodbc for secure parameterized queries



DATABASE DESIGN & RELATIONSHIPS

- **Core tables represent domain integrity and business rules. Referential Integrity enforces correct relationships.**

Main Tables

Users(user_id, role,contact)

Buses(busId, route, bus_number)

Seats (seat_number,is_booked)

PromoCodes(code, max_uses)

Waitlist(waitlist_id,position)

Key RelationShips:

One User → Many Bookings

One Bus→Many Waitlist entries

One Bus→ Many bookings

BusRes — Maheen (ADMIN)

 **BusRes**
Bus Reservation System

Maheen
ADMIN

Analytics

Revenue

Promo Codes

User Management

All Buses

Payment Methods

Route Finder

Sort & Stats

Admin Panel

Change Password

Logout

All Buses

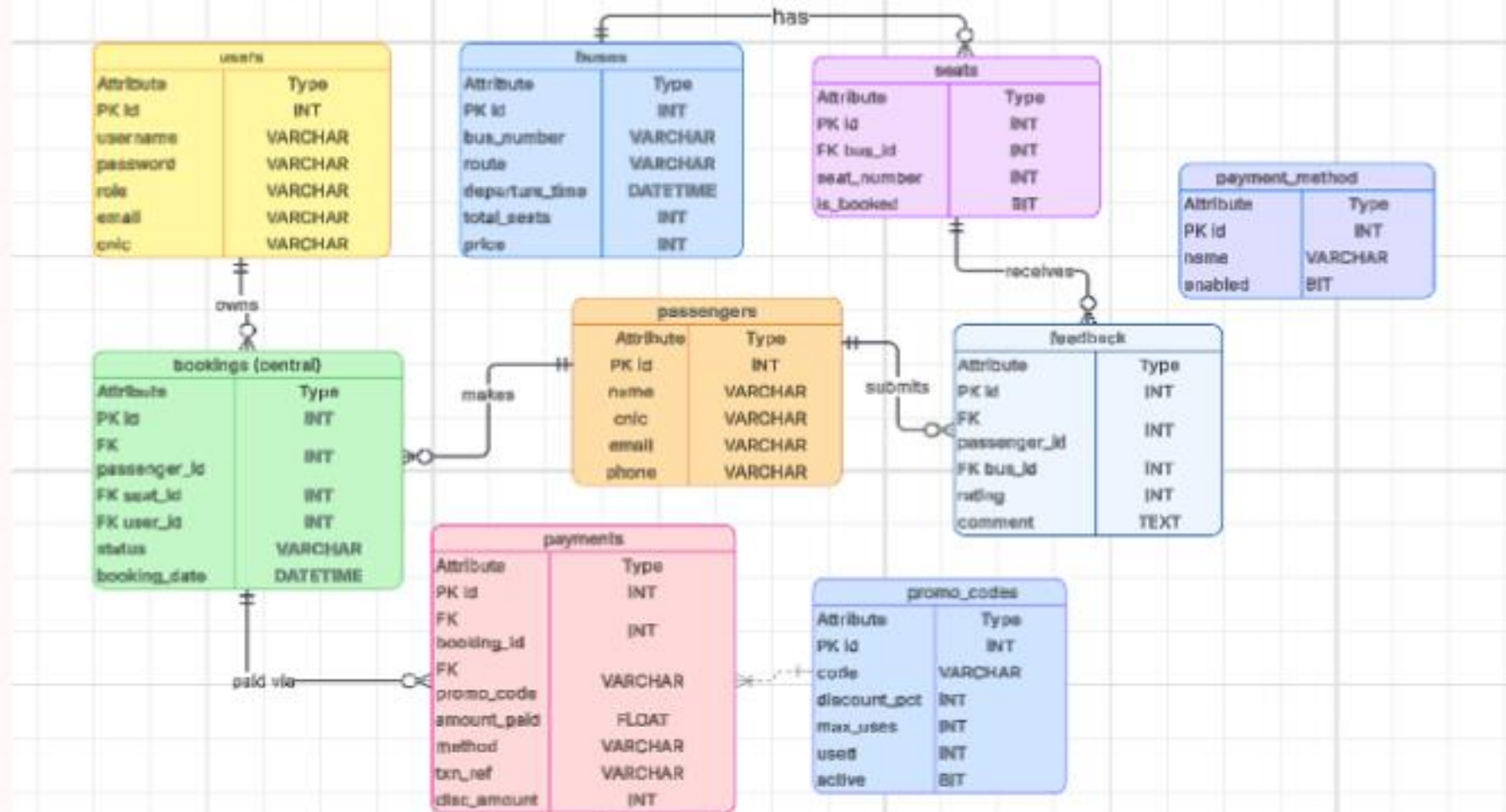
Read-only overview of all buses

All Columns

ID	Bus No	Route	Departure	Seats	Booked	Available
1	BUS-100	Islamabad to Rawalpindi	2026-06-03 02:37	42	42	0
2	BUS-101	Rawalpindi to Lahore	2026-06-10 23:27	41	41	0
3	BUS-102	Islamabad to Murree	2026-06-17 06:31	41	41	0
4	BUS-103	Peshawar to Islamabad	2026-06-13 19:03	40	8	32
5	BUS-104	Lahore to Multan	2026-06-08 15:13	9	9	0
6	BUS-105	Karachi to Hyderabad	2026-06-22 11:11	40	13	27
7	BUS-106	Quetta to Karachi	2026-05-29 04:34	40	8	32
8	BUS-107	Faisalabad to Lahore	2026-06-05 18:35	40	13	27
9	BUS-108	Multan to Bahawalpur	2026-05-28 08:04	40	12	28
10	BUS-109	Abbottabad to Islamabad	2026-06-15 23:17	40	9	31
11	BUS-110	Islamabad to Rawalpindi	2026-06-04 07:15	40	40	0
12	BUS-111	Rawalpindi to Lahore	2026-06-11 14:30	40	9	31
13	BUS-112	Islamabad to Murree	2026-06-18 09:00	40	11	29
14	BUS-113	Peshawar to Islamabad	2026-06-14 06:45	40	7	33
15	BUS-114	Lahore to Multan	2026-06-09 11:20	40	13	27
16	BUS-115	Karachi to Hyderabad	2026-06-23 16:00	40	5	35
17	BUS-116	Quetta to Karachi	2026-05-30 21:10	40	12	28
18	BUS-117	Faisalabad to Lahore	2026-06-06 08:50	40	40	0
19	BUS-118	Multan to Bahawalpur	2026-05-29 13:00	40	40	0

ER DIAGRAM

Bus Seat Reservation System ER Diagram



Entering System Application

